WORKED SOLUTION

STEP I 2000, Question 6

Pure • Curve sketching

STEP I 2000, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q6

The key idea

A quadratic expression in x and y that factorises into two linear factors describes a pair of straight lines. A product of two factors is positive exactly when the factors have the same sign, and negative when they have opposite signs, so each inequality picks out a pair of opposite "wedges" between two crossing lines. Once both regions are drawn on the same axes, finding a common point is a matter of looking.

Solution

The first factorisation and region

Expanding the right-hand side,

$$\begin{aligned}(x-y+2)(x+y-1) &= x^2+xy-x-xy-y^2+y+2x+2y-2\\ &= x^2-y^2+x+3y-2,\end{aligned}$$

as required. So the first inequality, after subtracting 2 from both sides, is

$$(x-y+2)(x+y-1)>0.$$

The boundary consists of the two lines

$$L_1:\ y=x+2, \qquad L_2:\ y=1-x,$$

which have gradients 1 and -1 and meet where $x+2=1-x$, that is at $(-\frac{1}{2},\frac{3}{2})$.

The product is positive when both factors have the same sign:

- both positive: $y<x+2$ and $y>1-x$, i.e. below L_1 and above L_2. This is the wedge to the *right* of the crossing point (it contains $(5,0)$, say);
- both negative: $y>x+2$ and $y<1-x$, i.e. above L_1 and below L_2. This is the wedge to the *left* of the crossing point (it contains $(-5,0)$).

So the region is the pair of opposite wedges, left and right, between the lines $y = x + 2$ and $y = 1 - x$, with the lines themselves excluded (on them the product is 0, not > 0). A quick test: at the origin the expression is $-2 < 0$, and the origin lies in the lower wedge, which is correctly excluded.

The second region

> **Ask yourself** How do we know this one factorises too, and what should the factors look like?

The quadratic terms $x^2 - 4y^2 = (x + 2y)(x - 2y)$ suggest trying

$$x^2 - 4y^2 + 3x - 2y + 2 = (x + 2y + a)(x - 2y + b).$$

The right-hand side expands to $x^2 - 4y^2 + (a + b)x + (2b - 2a)y + ab$. Matching coefficients gives $a + b = 3$ and $b - a = -1$, so $a = 2$, $b = 1$, and then the constant term $ab = 2$ also matches. Hence, adding 2 to both sides of the second inequality,

$$x^2 - 4y^2 + 3x - 2y < -2 \iff (x + 2y + 2)(x - 2y + 1) < 0.$$

The boundary lines are

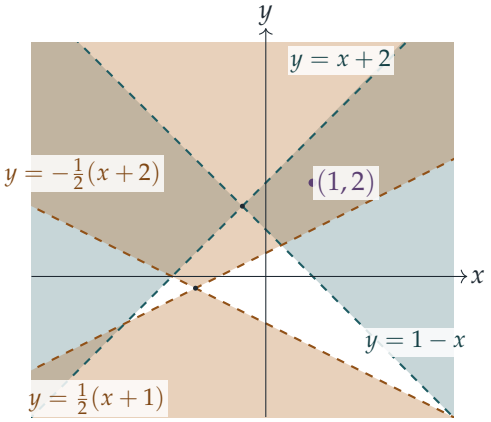
$$L_3 : \ y = -\tfrac{1}{2}(x + 2), \qquad L_4 : \ y = \tfrac{1}{2}(x + 1),$$

with gradients $-\frac{1}{2}$ and $\frac{1}{2}$, meeting where $-x - 2 = x + 1$, that is at $\left(-\frac{3}{2}, -\frac{1}{4}\right)$.

Now we need the factors to have *opposite* signs:

- $x + 2y + 2 > 0$ and $x - 2y + 1 < 0$: $y > -\frac{1}{2}(x + 2)$ and $y > \frac{1}{2}(x + 1)$, i.e. above both lines. This is the *upper* wedge (it contains $(0, 5)$);
- $x + 2y + 2 < 0$ and $x - 2y + 1 > 0$: below both lines. This is the *lower* wedge (it contains $(0, -5)$).

So the second region is the pair of opposite wedges, above and below, between the lines L_3 and L_4, again with the boundary excluded.



In the sketch the first region is shaded teal (left and right wedges), the second amber (upper and lower wedges), so the darker overlap is where both hold; the dashed lines are not included.

A point in both regions

The two pairs of wedges certainly overlap: the right-hand teal wedge opens out to the right between gradients ± 1, while the upper amber wedge contains everything above both of the shallower lines of gradient $\pm\frac{1}{2}$. For $x > -\frac{1}{2}$ the top edge $y = x + 2$ of the teal wedge rises faster than the line $y = \frac{1}{2}(x+1)$, so there is room between them.

Take, for instance, $(x, y) = (1, 2)$. Then

$$x^2 - y^2 + x + 3y = 1 - 4 + 1 + 6 = 4 > 2$$

and

$$x^2 - 4y^2 + 3x - 2y = 1 - 16 + 3 - 4 = -16 < -2,$$

so $(1, 2)$ satisfies both inequalities. (Checking against the factors: $(x - y + 2)(x + y - 1) = 1 \cdot 2 > 0$ and $(x + 2y + 2)(x - 2y + 1) = 7 \cdot (-2) < 0$.)

Checks

- Each factorisation can be checked at a random point: at $(3, 1)$, $x^2 - y^2 + x + 3y - 2 = 12$ and $(x - y + 2)(x + y - 1) = 4 \cdot 3 = 12$; at the same point $x^2 - 4y^2 + 3x - 2y + 2 = 14$ and $(x + 2y + 2)(x - 2y + 1) = 7 \cdot 2 = 14$.
- Test one point in each wedge, e.g. $(0, -5)$ gives $x^2 - 4y^2 + 3x - 2y = -90 < -2$, confirming the lower amber wedge belongs to the second region.
- A numerical check over 200 000 random points in $[-10, 10]^2$ found no disagreement between the inequalities and the wedge descriptions above.

Common slips

- Shading the wrong pair of wedges. Always test a point (the origin is the easiest) rather than relying on "above" and "below".
- Forgetting the -2 and $+2$ when moving the constants across: the first inequality is $(x - y + 2)(x + y - 1) > 0$, the second $(x + 2y + 2)(x - 2y + 1) < 0$.
- Drawing the boundary lines as solid: the inequalities are strict, so the lines are excluded.

Beyond STEP

A quadratic in two variables, $ax^2 + bxy + cy^2 + dx + ey + f$, defines a conic; when it factorises into two linear factors the conic is *degenerate*, a pair of lines, and this happens

exactly when a certain 3×3 determinant built from the coefficients vanishes. The sign pattern of a quadratic form (here $x^2 - y^2$ and $x^2 - 4y^2$, both indefinite) is studied in first-year linear algebra through symmetric matrices and their eigenvalues. Strang's *Introduction to Linear Algebra* is a good place to see this.

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